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Lab Exercises for Data Structure in C++

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LAB 1 : POINTERS

Exercise 1: Create a function to calculate (log) base-10 of user input number using pass-byreference method, display the original value and the result

Input compiler :

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4 void log10(double* num) {
5     double original = *num;
6     *num = log10(*num);
7     cout << "Original Value: " << original << endl;
8     cout << "Log10 Value: " << *num << endl;
9 }
10 int main() {
11     double number;
12     cout << "Enter a number: ";
13     cin >> number;
14     log10(&number);
15     return 0;
16 }
```

Output consol :

```
Enter a number: 1000
Original Value: 1000
Log10 Value: 3
```

Exercise 2: Create and array of 5 elements and display the addresses of all elements

Input compiler :

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int k[5] = { 10, 20, 30, 40, 50 };
5     int* ptr;
6
7     for (int i = 0; i < 5; i++) {
8         ptr = &k[i];
9         cout << "Address of k[" << i << "]: " << ptr << endl;
10    }
11    return 0;}
```

Output consol :

```
Address of k[0]: 00000019396FFB78
Address of k[1]: 00000019396FFB7C
Address of k[2]: 00000019396FFB80
Address of k[3]: 00000019396FFB84
Address of k[4]: 00000019396FFB88
```

LAB 2 : C++ Arrays

Exercise 1: Display the sum at average of array elements using Loop.

Input compiler :

```
1  #include<iostream>
2  using namespace std;
3
4  int main()
5  {
6      int array[5],x;
7      int sum=0;
8      cout << "enter 5 numbers :" << endl;
9      for(int i=0;i<5;i++)
10     {
11         cin >>x ;
12         array[i]=x;
13     }
14     for(int i=0;i<5;i++)
15     {
16         sum+=array[i];
17     }
18     cout << "the Average of sum is = " <<sum/5<< endl;
19
20     return 0;
21 }
22
```

Output consol :

```
enter 5 numbers :
10
20
30
40
50
the Average of sum is = 30
```

Exercise 2: use array to allow user print his name

Input compiler :

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main() {
6     string names[4];
7
8     for (int i = 0; i < 4; i++) {
9         cout << "Enter name " << i + 1 << ": ";
10        cin >> names[i];
11    }
12
13    cout << "The entered names are: ";
14    for (int i = 0; i < 4; i++) {
15        cout << names[i] << " ";
16    }
17    cout << endl;
18
19    return 0;
20 }
21
```

Output consol :

```
Enter name 1: ali
Enter name 2: mohamed
Enter name 3: saleh
Enter name 4: ahmed
The entered names are: ali mohamed saleh ahmed
```

LAB 3 : C++ STACK

Exercise 1: Create a Stack of size 10 , ask the user to push the elements, pop twice then print on screen the content of stack

Input compiler :

```
1  #include <iostream>
2  #include <stack>
3  using namespace std;
4
5  int main() {
6
7      const int SIZE = 10;
8      stack<int> stk;
9      int kamal = 1;
10     int element;
11
12     for (int i = 0; i < SIZE; i++) {
13         cout << "Enter push " << kamal << " : ";
14         cin >> element;
15         stk.push(element);
16         kamal++;
17     }
18
19     stk.pop();
20     stk.pop();
21
22     cout << "\nthe stack after popping two elements: ";
23     while (!stk.empty()) {
24         cout << stk.top() << " ";
25         stk.pop();
26     }
27
28     cout << endl;
29
30     return 0;
31 }
32
```

Output consol :

```
Enter push 1 : 10
Enter push 2 : 20
Enter push 3 : 30
Enter push 4 : 40
Enter push 5 : 50
Enter push 6 : 60
Enter push 7 : 70
Enter push 8 : 80
Enter push 9 : 90
Enter push 10 : 100
```

```
the stack after popping two elements: 80 70 60 50 40 30 20 10
```

Exercise 2: Modify a simple program that ask a user to enter numbers , then display only even numbers

Input compiler :

```
1 #include <iostream>
2 #include <stack>
3 using namespace std;
4
5 int main() {
6     stack<int> myStack;
7     int val;
8
9     cout << "Enter numbers: " << endl;
10    for (int i = 0; i < 5; i++) {
11        cin >> val;
12        if (val % 2 == 0) {
13            myStack.push(val);
14        }
15    }
16
17    cout << "The even numbers: " << endl;
18    while (!myStack.empty()) {
19        cout << myStack.top() << " ";
20        myStack.pop();
21    }
22    return 0;
23 }
24
```

Output consol :

```
Enter numbers:
1
7
10
12
14
The even numbers:
14 12 10
```

Exercise 3: Write a code that return true if the user write a character, and false if write an integer.

Input compiler :

```
1  #include <iostream>
2  #include <stack>
3  using namespace std;
4
5  int main() {
6      stack<string> Stack;
7      char input;
8
9      cout << "Enter a character or an integer: ";
10     cin >> input;
11
12     if (input >= '0' && input <= '9') {
13         Stack.push("false");
14     } else {
15         Stack.push("true");
16     }
17
18     while (!Stack.empty()) {
19         cout << Stack.top() << endl;
20         Stack.pop();
21     }
22     return 0;
23 }
24
```

Output consol :

```
Enter a character or an integer: h
true
```

LAB 4 : C++ Queue

Exercise 1: Create a Stack of size 10 , ask the user to push the elements, pop twice then print on screen the content of stack

Input compiler :

```
1  #include<iostream>
2  #include<queue>
3  using namespace std;
4
5  int main()
6  {
7      queue<int>qu;
8      int x,sum=0,z;
9      cout << "enter 5 elemnts plase: " << endl;
10     for(int i=0;i<5;i++)
11     {
12         cin >> x;
13         qu. push(x);
14         z=qu.size();
15     }
16
17     while(!qu.empty())
18     {
19         sum+=qu.front();
20         qu.pop();
21     }
22
23     cout<<"the Average is "<<sum/z<<endl;
24
25     return 0;
26 }
27
28
29
```

Output consol :

```
enter 5 elemnts plase:
100
200
300
400
500
the Average is 300
```

